

VISVESVARAYA TECHNOLOGICAL UNIVERSITY
“JNANA SANGAMA”, BELAGAVI -590 018.



DATABASE MANAGEMENT SYSTEMS (BCS402)

Mini-Project Report on

“Gym Membership Management System”

Submitted in partial fulfillment for the award of the degree of

BACHELOR OF ENGINEERING

In
CSE (AI & ML)

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CERTIFICATE

Certified that the mini-project work entitled **Gym Membership Management System** carried out by **VRISHABH P [1RN24CI231]**, **VRUNDA KULKARNI [1RN24CI232]**, **YAJNESH [1RN24CI233]**, **YASHAS S [1RN24CI234]** are Bonafide students of **RNS Institute of Technology** in partial fulfilment for the award of **“BACHELOR OF ENGINEERING”** in **CSE (AI&ML)** as prescribed by **VISVESVARAYA TECHNOLOGICAL UNIVERSITY, BELAGAVI** during the academic year **2025 – 26**. It is certified that all corrections/suggestions indicated for internal assessment have been incorporated in the report deposited in the departmental library. The mini-project report has been approved as it satisfies the academic requirements in respect of mini-project work on **DATABASE MANAGEMENT SYSTEMS (BCS402)** prescribed for the said degree.

Signature of Course

Coordinator Dept. of CSE

(AI&ML)

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ABSTRACT

The Gym Membership Management System is a database-driven application designed to efficiently manage and automate the day-to-day operations of a fitness center or gymnasium. In traditional gym environments, maintaining records of members, trainers, attendance, memberships, and payments manually can be time-consuming, error-prone, and difficult to manage as the number of members increases. The proposed system addresses these challenges by providing a centralized and organized database that stores all relevant information in a structured manner.

The system maintains comprehensive records of gym members, including personal details, membership plans, enrollment dates, attendance history, and payment information. It also stores trainer information and facilitates the assignment of trainers to members based on their fitness requirements. Membership plans of varying durations and costs can be created and managed efficiently, allowing administrators to monitor active and expired memberships. Additionally, the system records payment transactions and helps track membership renewals, ensuring smooth financial management within the gym.

The primary objective of this project is to reduce manual effort, improve operational efficiency, and enhance the accuracy of data management. By automating routine tasks such as member registration, membership renewal, trainer allocation, attendance tracking, and fee collection, the system minimizes human errors and eliminates redundant paperwork. The database design ensures data consistency, integrity, and security while enabling quick retrieval and updating of information whenever required.

The project is developed using the concepts of Database Management Systems (DBMS), including entity-relationship modeling, normalization, primary and foreign key constraints, relational schema design, and Structured Query Language (SQL). Various SQL operations such as insertion, deletion, updating, and retrieval of records are implemented to demonstrate effective database management techniques. The system also supports report generation, enabling administrators to obtain useful insights regarding memberships, payments, and member activities.

Overall, the Gym Membership Management System provides a reliable, scalable, and user-friendly solution for managing gym operations. It demonstrates the practical application of database technologies in solving real-world organizational problems and highlights the importance of efficient data management in modern fitness centers. The project serves as an excellent example of how DBMS concepts can be utilized to improve business processes, enhance productivity, and ensure effective information management.

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CHAPTER 1

INTRODUCTION

A Database Management System (DBMS) is software that enables users to create, manage, and manipulate databases efficiently. It provides mechanisms for data storage, retrieval, security, and consistency. DBMS plays a vital role in organizations that require systematic management of large volumes of data.

The Gym Membership Management System uses DBMS concepts to maintain gym-related information in a structured manner. The system stores member details, trainer information, membership plans, attendance records, and payment transactions in a centralized database.

1.1 Objectives

- Maintain member records digitally.
- Manage gym membership plans efficiently.
- Track membership payments and renewals.
- Maintain trainer information and assignments.
- Generate reports for administrative purposes.
- Reduce paperwork and manual errors.

1.2 Problem Statement

Managing gym operations manually can lead to data redundancy, errors in record keeping, and difficulties in tracking memberships and payments. A centralized database system is required to automate gym management activities and improve operational efficiency.

CHAPTER 2

PROJECT DESIGN

The Gym Membership Management System is designed to simplify gym administration by maintaining a centralized database of members, trainers, memberships, attendance, and payments. The system allows administrators to perform CRUD operations on records while ensuring data integrity and consistency.

2.1 Project Overview

The system manages the core functionalities of a gym, including member registration, membership management, trainer assignment, attendance tracking, and payment processing. The backend database stores all operational and transactional data.

2.2 Features

- ☐ Add, update, and delete member records.
- ☐ Manage membership plans and durations.
- ☐ Assign trainers to members.
- ☐ Track member attendance.
- ☐ Record and monitor payment transactions.
- ☐ Generate reports on memberships and payments.

2.3 ER Diagram

Entities:

Members

- Member_ID (PK)
- Name
- Gender
- Age
- Phone
- Address

Membership

- Membership_ID (PK)
- Plan_Name
- Duration
- Fee

Trainer

- Trainer_ID (PK)
- Trainer_Name
- Specialization
- Contact

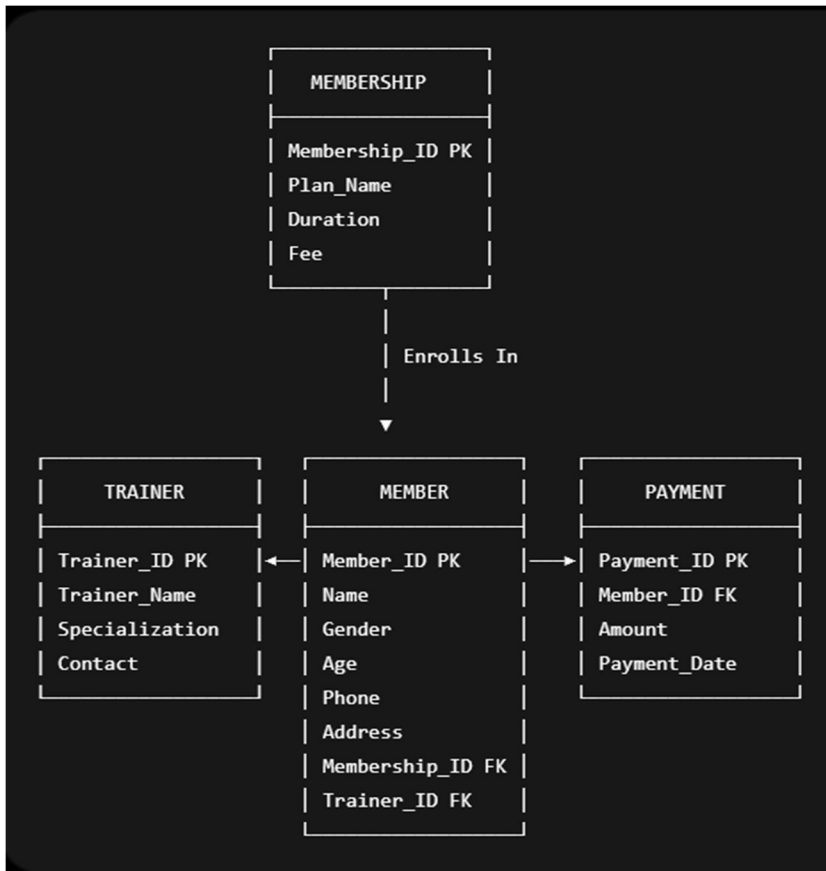
Payment

- Payment_ID (PK)
- Member_ID (FK)
- Amount
- Payment_Date

Relationships:

- Member enrolls in Membership.
- Member assigned to Trainer.
- Member makes Payment.

Diagram :



2.4 Technologies Used

- Database: MySQL
- Query Interface: MySQL Workbench

CHAPTER 3

IMPLEMENTATION

The implementation phase involves designing database tables, defining relationships, inserting sample data, and executing SQL queries to perform operations on gym records.

3.1 Schema Creation

```
CREATE TABLE Membership(  
Membership_ID INT PRIMARY KEY,  
Plan_Name VARCHAR(50),  
Duration VARCHAR(20),  
Fee DECIMAL(10,2)  
);
```

```
CREATE TABLE Trainer(  
Trainer_ID INT PRIMARY KEY,  
Trainer_Name VARCHAR(50),  
Specialization VARCHAR(50),  
Contact VARCHAR(15)  
);
```

```
CREATE TABLE Member(  
Member_ID INT PRIMARY KEY,  
Name VARCHAR(50),  
Gender VARCHAR(10),  
Age INT,  
Phone VARCHAR(15),  
Address VARCHAR(100),
```

```
Membership_ID INT,  
  
Trainer_ID INT,  
  
FOREIGN KEY(Membership_ID)  
  
REFERENCES Membership(Membership_ID),  
  
FOREIGN KEY(Trainer_ID)  
  
REFERENCES Trainer(Trainer_ID)  
  
);  
  
CREATE TABLE Payment(  
  
Payment_ID INT PRIMARY KEY,  
  
Member_ID INT,  
  
Amount DECIMAL(10,2),  
  
Payment_Date DATE,  
  
FOREIGN KEY(Member_ID)  
  
REFERENCES Member(Member_ID)  
  
);
```

3.2 Insert Data

3.2.1 Insert Membership Plans

```
INSERT INTO Membership  
  
VALUES  
  
(1,'Monthly','1 Month',1000),  
  
(2,'Quarterly','3 Months',2500),  
  
(3,'Yearly','12 Months',8000);
```

3.2.2 Insert Trainers

```
INSERT INTO Trainer
```

VALUES

(101,'Rahul Sharma','Weight Training','9876543210'),
(102,'Priya Singh','Cardio Fitness','9876501234');

3.2.3 Insert Members

INSERT INTO Member

VALUES

(201,'Amit Kumar','Male',24,'9876123456',
'Bangalore',1,101),
(202,'Sneha Rao','Female',22,'9876549876',
'Bangalore',2,102);

3.2.4 Insert Payments

INSERT INTO Payment

VALUES

(301,201,1000,'2025-09-01'),
(302,202,2500,'2025-09-02');

3.3 Queries for Operations

3.3.1 View All Members

SELECT * FROM Member;

3.3.2 View Membership Plans

SELECT * FROM Membership;

3.3.3 View Trainer Details

SELECT * FROM Trainer;

3.3.4 View Payment Records

SELECT M.Name,P.Amount,P.Payment_Date
FROM Member M

JOIN Payment P

ON M.Member_ID=P.Member_ID;

3.3.5 View Member and Assigned Trainer

SELECT M.Name,T.Trainer_Name

FROM Member M

JOIN Trainer T

ON M.Trainer_ID=T.Trainer_ID;

3.4 Update Operations

3.4.1 Update Member Phone Number

UPDATE Member

SET Phone='9988776655'

WHERE Member_ID=201;

3.4.2 Update Membership Fee

UPDATE Membership

SET Fee=9000

WHERE Membership_ID=3;

3.5 Delete Operations

3.5.1 Delete Payment Record

DELETE FROM Payment

WHERE Payment_ID=301;

3.5.2 Delete Member

DELETE FROM Member

WHERE Member_ID=202;

CHAPTER 4

RESULTS

The Gym Membership Management System successfully stores and manages information related to members, trainers, memberships, and payments. SQL queries were executed to perform insertion, retrieval, updating, and deletion of records. The system provides efficient management of gym operations while maintaining data consistency and integrity.

(Insert screenshots of table creation, data insertion, and query outputs here.)

FUTURE ENHANCEMENTS

- Online member registration.
 - Attendance tracking using QR codes.
 - SMS and email notifications.
 - Mobile application support.
 - Online payment gateway integration.
 - Automated membership renewal reminders.
-

CHAPTER 5

CONCLUSION

The Gym Membership Management System provides an efficient solution for managing gym operations through a centralized database. The project demonstrates the practical implementation of DBMS concepts such as relational schema design, normalization, primary keys, foreign keys, and SQL operations. The system reduces manual effort, improves data accuracy, and enhances the overall management of gym memberships and related activities.

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